



2nd Annual MidPoint Community Meetup

Try Me (Workshop): AI Assisted Integration of First Systems with MidPoint



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**[RECOVERY
AND RESILIENCE
PLAN]**

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Agenda

- Environment preparation
- Try me (workshop)
- Conclusion



*This presentation has been adapted from MCM 2026 session:
“Workshop: AI Assisted Integration of First Systems with MidPoint”.*

Environment Setup

- Install Docker
- Clone **midpoint-samples** repository
- Configure LLM gateway
- Create Docker containers



Environment: Install Docker

- Download and install [Docker for your operating system](#). You can install either:
 - Full **Docker Desktop** using the link above
 - (Linux-only) [Docker Engine](#) and [Docker Compose Plugin](#)
- **We recommend at least 16 GB RAM** as the environment contains multiple Docker containers
- **Please note:** TCP port 80 must be available on your machine

Environment: Clone midpoint-samples Repository (GitHub)

- Clone [midpoint-samples](#) repository
- Go to `samples/demo/smart-resource-integration` directory
- Docker configuration is in `docker/smart-resource-integration` subdirectory

Environment: Configure LLM Gateway

- Configure **midpilot-smart-integration** container to use your LLM gateway in `docker-compose.yml` file:
- OpenRouter hosted-LLM example (other gateways including self-hosted ones are possible)

```
midpilot-smart-integration:
```

```
  . . .
```

```
  environment:
```

```
    LOGGING__LEVEL: info
```

```
    LLM__OPENAI_API_BASE: https://openrouter.ai/api/v1
```

```
    LLM__MODEL_NAME: openai/gpt-oss-120b
```

```
    LLM__EXTRA_BODY: "{\"provider\":{\"sort\":\"throughput\"}}"
```

```
    LLM__OPENAI_API_KEY: ${OPENAI_API_KEY:-}
```

```
    # enable langfuse for debugging purposes
```

```
    LANGFUSE__TRACING_ENABLED: false
```

```
  networks:
```

```
    - net
```

Environment: Specify API Key for LLM Gateway

- Create API key/token for LLM gateway
- Configure environment variable `OPENAI_API_KEY` for Docker containers: `.env` file
 - **midpilot-smart-integration** container uses the variable

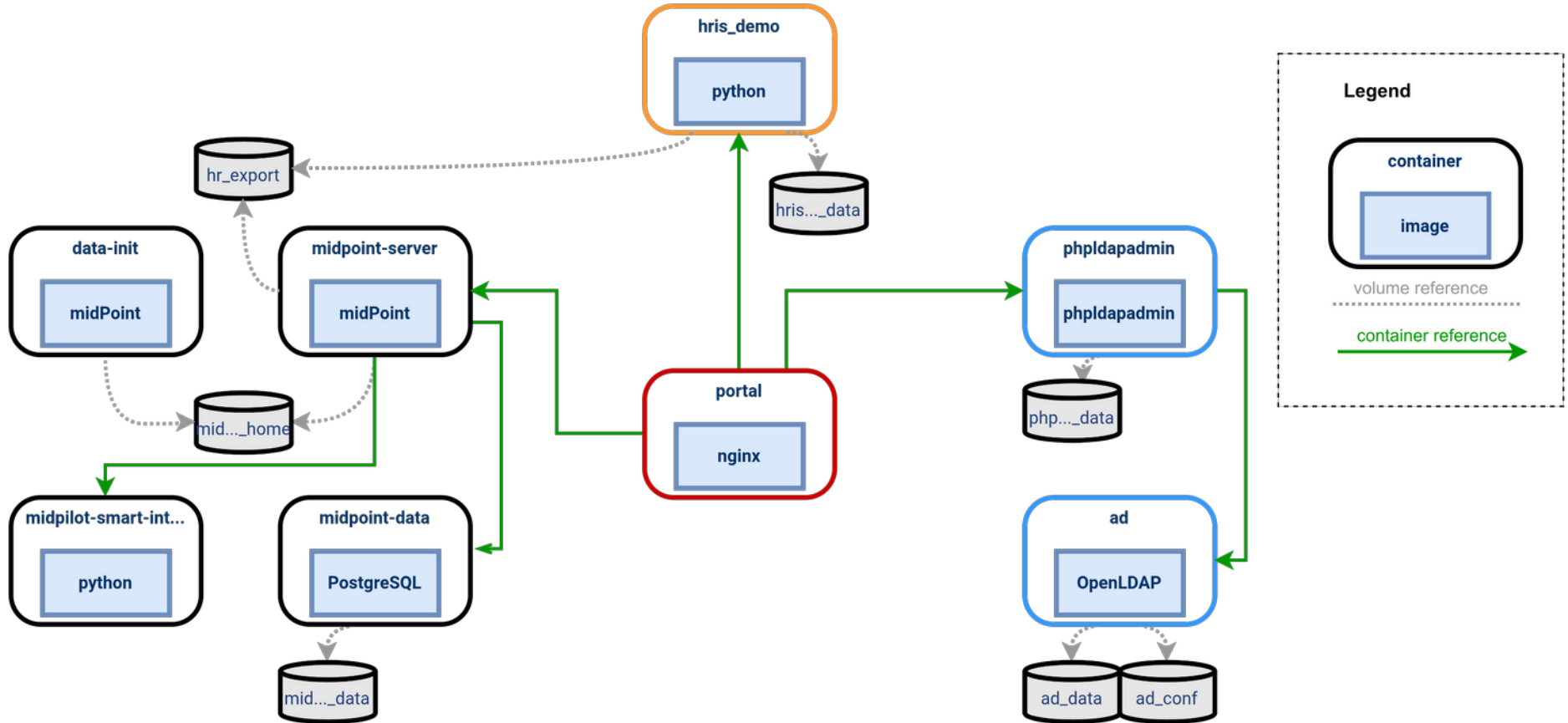
```
# These parameters can be overridden by setting environment variables before calling docker-  
compose up  
# MidPoint-related  
ENV=demo  
USERTOKEN=  
REPO_MISSING_SCHEMA_ACTION=create  
REPO_UPGRADEABLE_SCHEMA_ACTION=stop  
MEM_MAX=4096m  
MEM_INIT=2048m  
TZ=UTC  
#MP_VER=4.9  
# OPENAI_API_KEY for workshop usage (MCM 2026). Replace with a real value.  
OPENAI_API_KEY=sk-or-v1-abc...123
```

Environment: Create and Start Docker Environment

- All Docker containers can be created from publicly available Docker images
 - Note: the Docker images were updated since MCM 2026
- Run “docker compose up -d”

```
cd smart-resource-integration  
docker compose up -d
```


Environment Docker Architecture



AI Service Health Check

Portal page: <http://localhost>

Welcome to midPoint demo!

Please choose where you want to go (all links open in new window/tab).

midPoint



HRIS

php
LDAP
admin



Demo [cheatsheet](#).

Pretty-print ☒

```
{
  "status": "OK",
  "version": "0.1.0 (469ed4b)",
  "checks": [
    {
      "name": "server",
      "status": "OK",
      "error": null
    },
    {
      "name": "llm",
      "status": "OK",
      "error": null
    },
    {
      "name": "langfuse",
      "status": "DISABLED",
      "error": null
    }
  ]
}
```

Log In to MidPoint

Portal page: <http://localhost>

Welcome to midPoint demo!

Please choose where you want to go (all links open in new window/tab).



HRIS



Demo [cheatsheet](#).

Evolveum

midPoint

Login to your account

Please enter account details to start the session

Username

Password

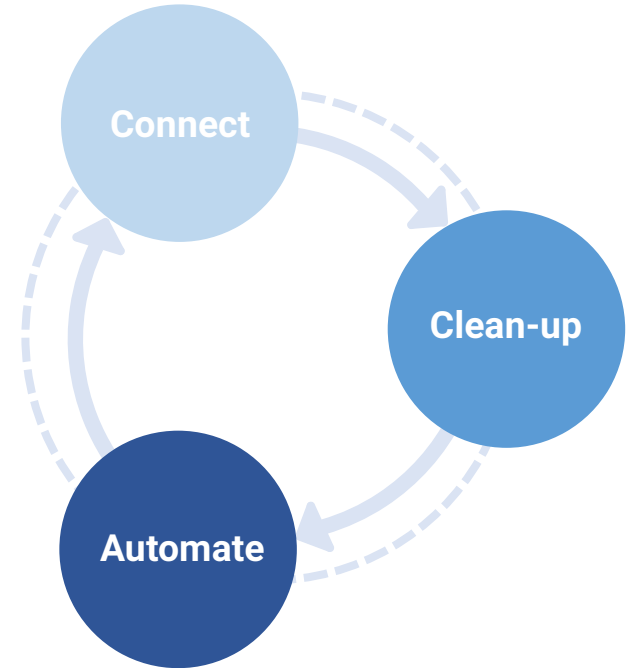
Sign in

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No active subscription. Please support midPoint by purchasing a subscription.

English <

First Steps Methodology

- *Iterative* identity management program (Est. 2023)
- Guides you through the quick midPoint deployment in three steps
- **Connect, Clean-up, Automate**
- Docs: [First Steps Methodology](#) (expect updates for midPoint 4.11)



How First Steps Methodology Works

- **Goal: start using midPoint** in your organization by connecting your first source and target systems
- **Simple** – GUI wizards, no XML
- **Safe** – prevents unexpected changes or deletions of existing data by using simulations and marks (exceptions)
- **Configuration assistants (AI/heuristics)** – enable administrators to focus on data and business requirements rather than midPoint internals **NEW | MIDPILOT**
- **MidPilot:** AI-powered assistant designed to accelerate application onboarding into midPoint (generate connector code, recommend attribute mappings and correlations, and more)



Why Simulations?

- Configuration is seldom correct for the first time
- Data can be unexpectedly deleted or modified by incorrect configuration or if target system data is inconsistent
- Incremental adding of new features is **safer** with simulations
- “**What would happen if...**” → You have time to react before you damage anything
- Clever usage of “lifecycle state” and “execution mode” to perform “Development” and “Production” simulation
- Docs: [Simulations](#)



(Teaser) Target System Reconciliation Simulation

Simulation result

← Back to object type wizard

Simulation task details

Start timestamp

May 4, 2026, 3:59:19 PM UTC

End timestamp

May 4, 2026, 3:59:22 PM UTC

Finished in

2 seconds

Task

Preview of Reconciliation: LDAP: User accounts

Status

Finished

Configuration

Development

Added objects

0

Deleted objects

0

Modified objects

10

Unmodified objects

104

Event marks

Focus activated

0

Focus archetype changed

0

Projection deactivated

0

Projection password changed

0

Projection renamed

0

Projection identifier changed

0

Projection entitlement changed

0

Resource object affected

5

More info

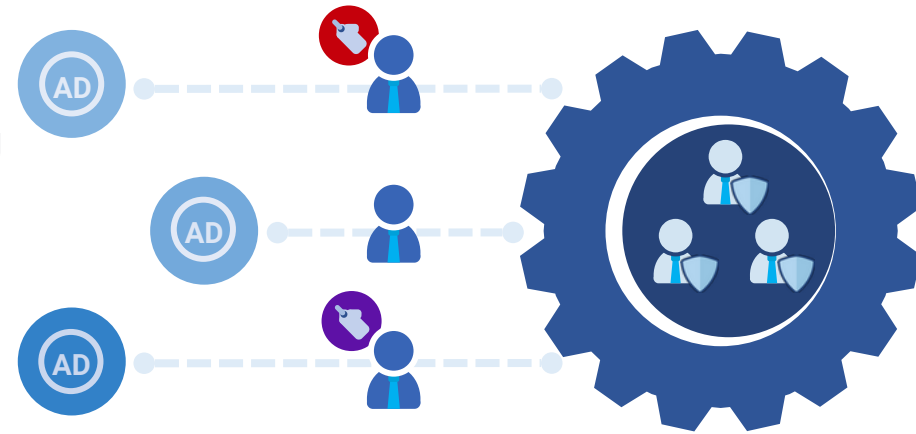
Modify Shadow uid=afreeman,ou=users,dc=example,dc=com

Modify User accounts attributes

Item	Old value	New value
title	Sales Rep.	Sales Representative (Example, Inc.)
cn	Alex Freeman	Alexander Freeman
givenName	Alex	Alexander
displayName	Alex Freeman	Alexander Freeman

Why (Account) Marks?

- **Marks** allow **exceptions** from policies
 - Protected, Do not touch, Correlate later and more
- Example: *permanently* protect certain accounts from being processed or deleted (e.g., system accounts)
- Example: *temporarily* protect certain accounts from being processed or deleted, and postpone the decision to a later time to allow the deployment process to continue (e.g., accounts that seem to be system accounts)
- Also useful in reporting and analytics



First Steps Methodology Overview

1 Plan Your Deployment

Identify data sources, targets, plan resources, timing and money

2 Connect the First Source System (HR)

Connect HR system and preview data

3 Import Source Data

Import data from HR system, create users in midPoint

4 Connect the First Target System

Connect the most important target system and preview data

5 Target System Integration

Correlate existing accounts to midPoint users

6 (Opt.) Import Usernames from Target System

Import usernames from the first target system to midPoint

7 Enable Provisioning to Target System

Configure Target for provisioning from midPoint, first with simulation

8 Automate Integration

Automate Source→Target account provisioning (JML)

9 Override Incorrect Data

Make sure we can override incorrect data from HR if needed

Try me (Workshop)

Please ensure you have configured the environment.

AI results may vary. *The outputs generated by AI/Large Language Models (LLMs) are probabilistic and may vary in time, even with identical prompts. You may need to re-generate the results multiple times.*

1. Plan Your Deployment (Kick-off) – Identify Source and Target Systems

Source system: HRIS export (CSV); employees + contractors (emptype)
ID: empnum (employee number)

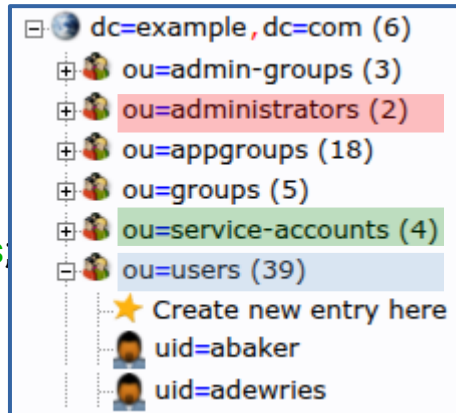
empnum	firstname	surname	emptype	job	status	locality
1001	Geena	Green	FTE	124#CEO	Active	Hot Lava City
1002	Ana	Lopez	FTE	125#CFO	Active	Hot Lava City
1003	Jimmy	Taylor	FTE	107#Junior Consultant	Former employee	Small Red Rock City
1004	Peter	Hunter	FTE	910#HR Consultant	Active	White Stone City
1016	Jane	Anderson	FTE	107#Junior Consultant	Long-term leave	Hot Lava City
8000	Janet	Garner	CONTRACTOR	899#Cleaning & Maintenance Specialist	Active	Hot Lava City
8001	Ben	Goosehead	CONTRACTOR	899#Cleaning & Maintenance Specialist	Active	Hot Lava City
8002	Maria	Alvarez	CONTRACTOR	899#Cleaning & Maintenance Specialist	Active	Small Red Rock City

Fixed set

Prefix

Fixed set

Target system: corporate directory server (LDAP);
standard accounts + admin accounts + service accounts;
internal groups + application groups + admin groups
Account ID: uid



2. Connect the First Source System (HR)

- Connect Source system using a CSV file and preview resource data
- Classify data from resource object classes into object types (which define the appearance and behavior of resource objects)
 - Try me (workshop): use “emptytype” column
- Configure object type for “Employees”
 - Synchronization (“Source” system)
 - Inbound mappings
- Use a series of wizards and configuration assistants

2. Connect the First Source System (HR) – Object Types (Expected Result)

HR / Object types / [Review suggestions](#)

Select object type to review

Review the suggested object types generated based on your resource data. Select the one that best matches the kind of objects you want to manage. You can discard and refresh to possibly find new suggestions to review.

Full-time employee accounts

☒ Accounts where c:attributes/ri:emptytype indicates a full-time employee (FTE). This attribute is stable and uniquely identifies the employment type of the account holder, making it suitable for partitioning. [Hide filter](#) ^ ...

Kind: account Intent: employee Object class: AccountObjectClass Focus type: UserType

Filter:
c:attributes/ri:emptytype = "FTE" [AI](#)

Contractor accounts

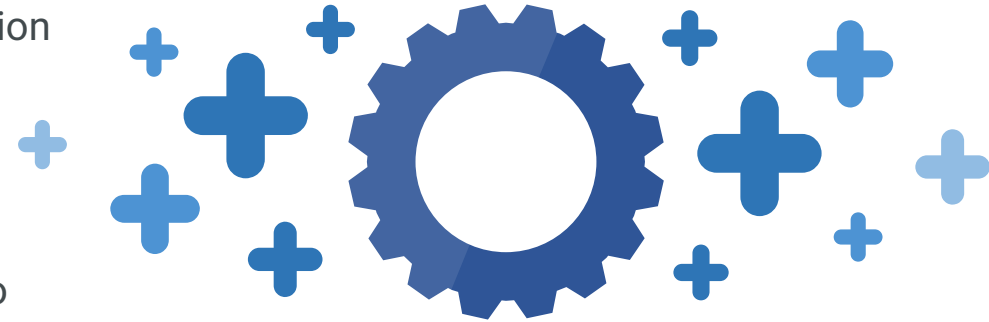
☐ Accounts where c:attributes/ri:emptytype indicates a contractor (CONTRACTOR). This stable attribute separates contractor accounts from full-time employees. [Show filter](#) v ...

Kind: account Intent: contractor Object class: AccountObjectClass Focus type: UserType

[Exit suggestions](#) [Refresh suggestions](#) [Review selected](#)

3. Import Source Data

- Simulate source data import to validate the configuration (especially mappings)
- Import data from Source system and create users in midPoint
- Inspect created users (properties, “Projection” – link to Source system account, archetype, state)

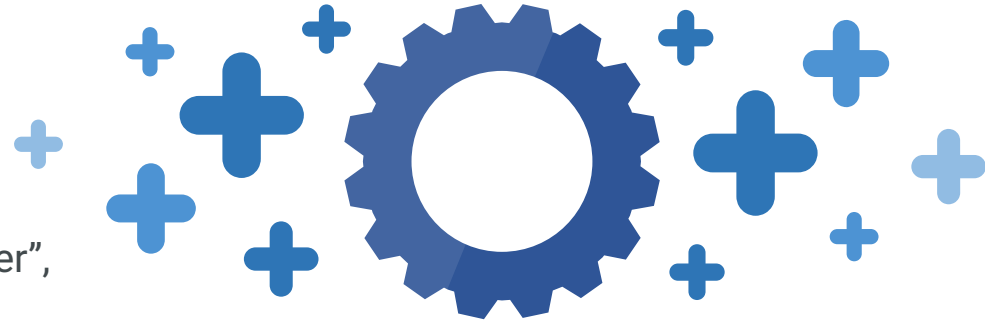


4. Connect the First Target System (LDAP)

- Connect Target system using LDAP connector and preview resource data
- Classify data from resource object classes into object types (which define the appearance and behavior of resource objects)
 - Try me (workshop): entry position in LDAP tree (subtree)
- Configure object type for “Employee accounts”
 - Correlation – by personalNumber
 - Synchronization (“Target” system)
 - Simulate the correlation
- Use a series of wizards and configuration assistants

5. Target System Integration

- Simulate reconciliation
- There may be uncorrelatable data:
 - Orphaned accounts
 - System (service) accounts
 - Unmatched accounts with invalid correlation data
- Mark uncorrelatable data as exceptions (“Correlate later”, “Do not touch” etc.) and plan to handle them later
- Correlate & link existing accounts to midPoint users
- Majority of the target system accounts are now linked to their midPoint owners



6. (Optional) Import Usernames from Target System

- You can import usernames to midPoint (if HR system does not have them)
- Reconfigure the original mapping in Source resource to be “weak” (E.g.: empnum → name)
- Prepare a new inbound mapping from Target system
- Simulate username import from Target system to midPoint
- Import usernames from Target system to midPoint

☐		abaker	1021	Alice Baker
☐		adewries	1030	Amanda de Wries
☐		afreeman	1010	Alexander Freeman
☐		ajackson	1029	Ashley Jackson
☐		alopez	1002	Ana Lopez
☐		bcarpenter	1024	Brad Carpenter
☐		cwhitehe	1039	Charles Whitehead
☐		ddavis	1007	Diane Davis
☐		diverson	1022	David Iverson

7. Enable Provisioning to Target System

- Configure Target resource for provisioning from midPoint (Outbound Mappings + Activation, Credentials) using AI/heuristics configuration assistants **NEW | MIDPILOT**
- Set all outbound mappings' lifecycle state to: Proposed
- Simulate reconciliation, review data inconsistencies
 - Update mappings, use marks or allow midPoint to update target system data
- Set all outbound mappings' lifecycle state to: Active
- Run reconciliation with Target System



8. Automate Integration

- Automate Source → Target provisioning (Joiners – Movers - Leavers)
- Deactivate username inbound mappings in both resources
- Turn on (or customize) the username generator in midPoint for all Person users – in Person Object Template
- Add inducement to create Target system account for all Person users – in Person Object Template
- **Suspend the legacy on-boarding/provisioning processes for Target system**
- Schedule regular reconciliation with Source system
- Person archetype aggregates all birthright accesses for new users



Conclusion

- First Steps methodology: integrate the first source and target systems (consider attending **Updated MidPoint Methodology session** for more information)
- **Configuration assistants** from midPilot project
- **Iterative, safe** approach using **simulations** and **marks**
- Gradual migration to support business continuity
- The current **MidPoint Deployment: First Steps self-paced training** is [available free of charge for everyone](#)
- Trainings will be updated for midPoint 4.11
- See how you can [access our learning portal](#)



Stop Docker Environment

- Stop (but keep) the Docker environment for later use

```
cd smart-resource-integration  
docker compose down
```

Stop and Remove Docker Environment

- Stop and remove the Docker environment (containers, volumes, networks)

```
cd smart-resource-integration  
docker compose down -v
```



Thank you for your attention

Feel free to ask your questions now!



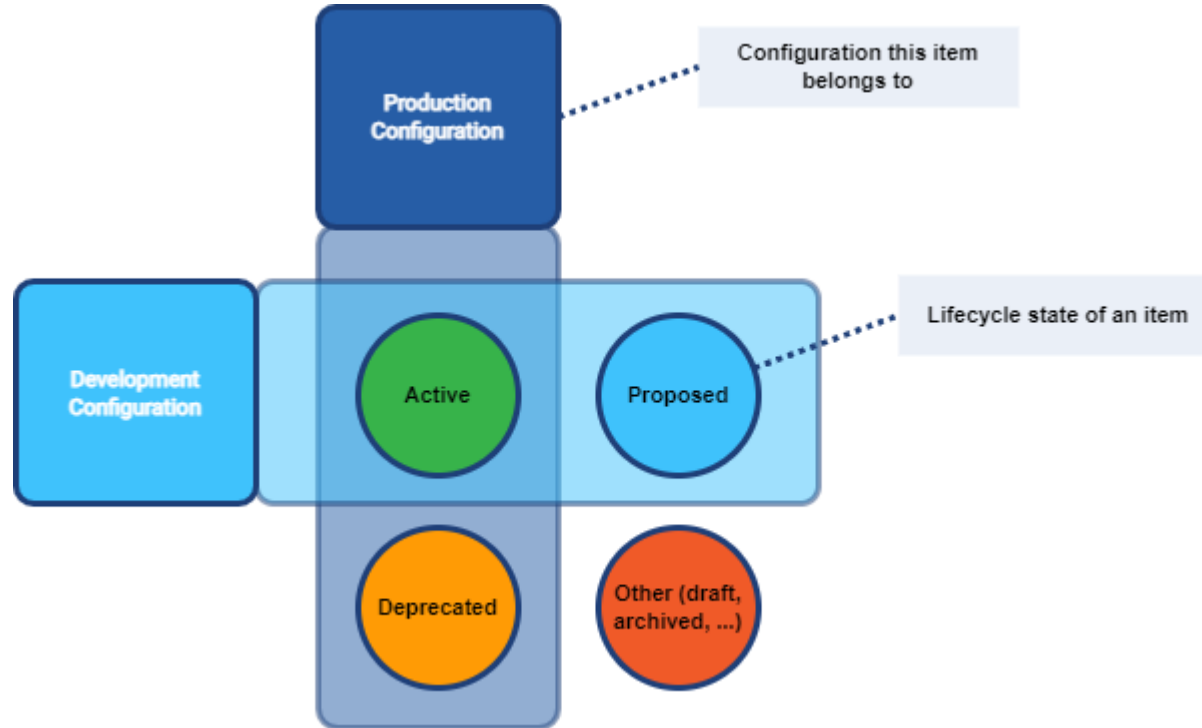
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BONUS: Configuration Item Lifecycle and System Configurations



Docs: [Configuration Item Lifecycle and system Configurations](#)